Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2000-I-Q05] | Part A (Q1-Q5)

Two boxes each contain black and white marbles, with 25 marbles altogether. One marble is selected uniformly from each box, independently. Both are black with probability $27/50$. If both-white probability is m/n reduced, find $m+n$.

两盒中均有黑白两色球，总数为 25。独立从每盒等概率取一球，两球均黑的概率为 $27/50$。若两球均白的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2000-II-Q03] | Part A (Q1-Q5)

A deck has four copies of each label 1 through 10. Remove two equal-labeled cards, then uniformly choose two of the remaining cards without replacement. If their matching probability is m/n reduced, find $m+n$.

一副牌有 1 至 10 各四张。先移去两张同号牌，再从余牌中不放回等概率选两张。若两张同号的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2002-I-Q01] | Part A (Q1-Q5)

A uniformly random plate consists of three letters from a 26-letter alphabet followed by three decimal digits, with repetitions allowed. If at least one of the two three-symbol blocks is palindromic with probability m/n reduced, find $m+n$.

车牌由三个二十六字母及三个十进制数字依次组成，允许重复，所有排列等可能。若至少一个三符号块为回文的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2005-II-Q02] | Part A (Q1-Q5)

Three rolls of each of three types are randomly distributed into three labeled bags, three rolls per bag. The probability every bag has all three types is m/n in lowest terms. Find $m+n$.

三种面包各三个，随机装入三个有标号的袋子，每袋三个。每袋恰含三种的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2010-I-Q01] | Part A (Q1-Q5)

Choose two distinct positive divisors of 2010^2 uniformly. If the probability that exactly one is a square is m/n in lowest terms, find $m+n$.

从 2010^2 的正约数中等可能选出两个不同的数，恰有一个为平方数的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2010-II-Q04] | Part A (Q1-Q5)

Twelve airport gates lie in a row, with adjacent gates 100 feet apart. Select an old gate uniformly and then a different new gate uniformly. The probability that the walk is at most 400 feet is m/n in lowest terms. Find $m+n$.

十二个机场登机口排成一行，相邻间距一百英尺。先等可能选原登机口，再从其余口等可能选新口。步行不超过四百英尺的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2013-I-Q04] | Part A (Q1-Q5)

A thirteen-cell square array has quarter-turn symmetry about its central cell. The other twelve cells form three four-cell rotation orbits. Color eight cells red and five blue, uniformly among all choices. If the chance of a quarter-turn-invariant coloring is $1/n$, find n.

一个十三方格阵列关于中心方格具有九十度旋转对称，其余十二格分为三个各含四格的旋转轨道。等可能将八格涂红、五格涂蓝。若染色仍具有此旋转对称的概率为 $1/n$，求 n。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2015-I-Q02] | Part A (Q1-Q5)

Nine people are split into nationality groups of sizes $2, 3, 4$. A uniformly random three-person subset is chosen. The probability its group-size pattern is exactly $2 + 1$ is reduced m/n. Find $m + n$.

九人按国籍分成 $2, 3, 4$ 人三组，等可能选出三人。所选三人的国籍人数分布恰为 $2 + 1$ 的概率是最简分数 m/n。求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2015-II-Q05] | Part A (Q1-Q5)

Two different cells are chosen uniformly from an $n \times n$ grid, where $n \geq 2$. Find the smallest n such that the probability they share a full side is less than $1/2015$.

从 $n \times n$ 方格中等可能选两个不同小格，$n \geq 2$。求使两格共边概率小于 $1/2015$ 的最小 n。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2018-I-Q03] | Part A (Q1-Q5)

Shuffle five red and five green cards, then expose the first five in order. Happiness means all exposed reds form one block and all exposed greens form one block, allowing one color to be absent. Reduce this probability m/n and find $m+n$.

将五红五绿牌洗匀，依次翻出前五张。若翻出的红牌各自连续、绿牌也各自连续（允许某色缺席），称为成功。将概率约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2019-I-Q02] | Part A (Q1-Q5)

Choose J uniformly from one through twenty, then B uniformly from the nineteen other integers. If $P(B - J \geq 2) = m/n$ reduced, find $m + n$.

先从一至二十等可能选 J，再从其余十九数选 B。若 $P(B - J \geq 2) = m/n$ 最简，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2019-I-Q05] | Part A (Q1-Q5)

From $(4,4)$ a particle independently takes one of $(-1,0),(0,-1),(-1,-1)$ with probability one third, stopping at its first axis point. If the probability it stops at the origin is $m/3^n$ with $3\nmid m$, find $m+n$.

粒子从 $(4,4)$ 出发，每步独立等可能选左、下、左下一个单位，首次到坐标轴即停止。在原点停下的概率为 $m/3^n$，m 不被三整除，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2023-I-Q01] | Part A (Q1-Q5)

Place five men and nine women uniformly at random in fourteen evenly spaced positions on a circle. Let m/n be the reduced probability that no man faces another man across a diameter. Compute $m+n$.

五名男子与九名女子随机等可能地站在圆周上十四个等距位置。任意男子的正对面均为女子的概率约为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2026-II-Q05] | Part A (Q1-Q5)

An urn has r red and b blue marbles, with $r, b \geq 7$. Drawing seven without replacement, the probabilities of exactly four red and exactly five red are equal. Sum the five least possible totals $n = r + b$.

袋中有红球 r 个、蓝球 b 个，均不少于七。无放回抽七个，恰有四红与恰有五红的概率相等。求最小的五个可能总数 $n = r + b$ 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q06] | Part B (Q6-Q10)

Roll a fair six-sided die four times. The probability that the four results are nondecreasing is m/n reduced. Find $m+n$.

公平六面骰掷四次，四个结果非降的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q10] | Part B (Q6-Q10)

Uniformly choose two distinct points from the lattice box $0 \le x \le 2, 0 \le y \le 3, 0 \le z \le 4$. If their midpoint is also in the box with probability m/n reduced, find $m+n$.

从整点盒 $0 \le x \le 2, 0 \le y \le 3, 0 \le z \le 4$ 中等概率选两个不同点。若中点仍属该整点盒的概率约分为 m/n，求 $m+n$。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-II-Q09] | Part B (Q6-Q10)

Independently color each cell of a 3×3 board red or blue with equal probability. If the probability of no all-red 2×2 block is m/n reduced, find $m+n$.

将 3×3 方格的每格独立等概率染红或蓝。若不存在全红 2×2 方块的概率约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2004-II-Q10] | Part B (Q6-Q10)

Choose uniformly among positive integers at most 2^{40} with exactly two binary ones. If the chance of divisibility by 9 is p/q reduced, find $p+q$.

从不超过 2^{40} 且二进制恰有两个一的正整数中等概率选一个。若被九整除的概率约分为 p/q，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2005-I-Q09] | Part B (Q6-Q10)

Twenty-seven cubes each have two adjacent unpainted faces and four orange faces. Random independent orientations form a $3 \times 3 \times 3$ cube. Its all-orange exterior has probability $p^a/(q^b r^c)$ in prime-factor form. Find $a+b+c+p+q+r$.

二十七个小立方体各有两个相邻面未染色，其余四面为橙色。随机独立定向后拼成 $3 \times 3 \times 3$ 大立方体。外表全橙的概率分解为 $p^a/(q^b r^c)$，求 $a+b+c+p+q+r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2007-II-Q10] | Part B (Q6-Q10)

Independently choose uniform random subsets A, B of a six-element set S. The probability that $B \subseteq A$ or $B \subseteq S \setminus A$ is m/n^r with prime n and reduced numerator. Find $m+n+r$.

在六元素集合 S 中独立等可能选子集 A, B。事件 $B \subseteq A$ 或 $B \subseteq S \setminus A$ 的概率写成最简的 m/n^r，n 为素数，求 $m+n+r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2011-I-Q10] | Part B (Q6-Q10)

Three distinct vertices of a regular n-gon are chosen uniformly. The probability that they form an obtuse triangle is $93/125$. Find the sum of all possible n.

从正 n 边形顶点中等可能选三个不同顶点，构成钝角三角形的概率为 $93/125$。求所有可能 n 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2013-I-Q06] | Part B (Q6-Q10)

Twelve distinct books, including three mathematics books, are randomly packed into boxes of capacities three, four, five. The probability that the three math books occupy one box is reduced m/n. Find $m+n$.

十二本不同书中有三本数学书，随机装入容量分别为三、四、五本的三个盒子。三本数学书在同盒的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2020-I-Q09] | Part B (Q6-Q10)

Select three positive divisors of 20^9 independently and uniformly, calling them a_1, a_2, a_3. The chance that $a_1 \mid a_2$ and $a_2 \mid a_3$ is m/n reduced. Find m.

从 20^9 的正约数中独立等概率选取 a_1, a_2, a_3。若 $a_1 \mid a_2 \mid a_3$ 的概率约分为 m/n，求 m。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2022-I-Q09] | Part B (Q6-Q10)

Twelve blocks consist of two copies of each of six colors. A uniformly random row is formed. What is $m+n$ if the probability that every same-color pair has an even number of blocks between its copies is m/n in lowest terms?

六种颜色各有两块，共十二块，随机等可能排成一行。若每对同色块之间的块数均为偶数的概率为最简分数 m/n，求 $m+n$。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2025-I-Q07] | Part B (Q6-Q10)

Randomly pair letters A through L, alphabetize each pair, then sort the six two-letter words. If the probability that the last word contains G is m/n in lowest terms, determine $m+n$.

将字母 A 至 L 随机配成六对，每对按字母顺序排列，再把六个双字母词排序。最后一个词含 G 的概率为最简分数 m/n，求 $m+n$。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q15] | Part C (Q11-Q15)

Uniformly label the eight faces of a regular octahedron by 1 through 8. Consecutive labels, including 8 and 1, must not lie on edge-sharing faces. If the probability is m/n reduced, find $m+n$.

将 1 至 8 等概率标在正八面体八个面上。要求循环相邻标号（包括 $8,1$）不能位于共边面上。若概率约分为 m/n，求 $m+n$。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2011-II-Q12] | Part C (Q11-Q15)

Nine delegates, three from each of three countries, sit uniformly at random in nine labeled chairs around a circle. The probability that everyone has at least one neighbor from another country is reduced m/n. Find $m+n$.

三国各三名代表随机等可能地坐在环形九个固定座位上。每人至少有一名邻座来自其他国家的概率为最简分数 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2014-I-Q12] | Part C (Q11-Q15)

Choose f, g independently and uniformly from all functions on a four-element set to itself. The probability that their image sets are disjoint is reduced m/n. Find m.

从四元集合到自身的全部函数中，独立等可能选 f, g。两者像集不相交的概率为最简分数 m/n，求 m。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2014-II-Q13] | Part C (Q11-Q15)

Ten people contribute one left and one right shoe each. Pair all left shoes bijectively with right shoes uniformly. Require that no selection of k resulting pairs contains exactly the shoes of k owners for any $1 \leq k < 5$. Reduce the probability to m/n and find $m+n$.

十人各提供一左一右鞋，将十左鞋与十右鞋随机等可能一一配对。要求任取 k 双配成鞋都不能恰好包含 k 人的全部鞋，$1 \leq k < 5$。将概率约为 m/n，求 $m+n$。

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Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2018-I-Q12] | Part C (Q11-Q15)

Choose a subset uniformly from $\{1, \ldots, 18\}$. If the probability its element sum is divisible by three is reduced m/n, find m. The empty sum is zero.

从一至十八的所有子集中等可能选一个，空集和为零。若元素和可被三整除的概率为最简分数 m/n，求 m。

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